

Bangabandhu Sheikh Mujibur Rahman Science and Technology University
Department of Computer Science and Engineering (ICT)
4th Year 2nd Semester B.Sc. Engg. Examination 2022

Course No.: CSE457

Course Title: Cryptography and Network Security

Full Marks: 60

Time: 3 (Three) Hours

N.B.:

- 1) Answer **SIX** questions taking any **THREE** from each section.
- 2) The figures in the right margin indicate full marks.
- 3) Sequence must be maintained for each question.

SECTION-A (30 Marks)

1. a) What is cryptography and the application of cryptography. What are the characteristics of good cipher. 4
b) Describe the RSA based decryption mixnet. 6
2. a) Describe the classification of cryptosystems. 4
b) What are the requirements of blind signature? 4
c) Write down the advantage and disadvantage of blind signature? 2
3. a) Encrypt the message "the house is being sold tonight" using Affine cipher with key = (15, 20). 4
Ignore the space between words. Decrypt the ciphertext to get plaintext.
b) The encryption key in a transposition cipher is (3, 2, 6, 1, 5, 4). Find the decryption key. 3
c) Using Chinese Remainder Theorem, find an integer that has a remainder of 3 when divided by 7 and 13, but is divisible by 12. 3
4. a) Describe the Hwang's Scheme. 5
b) Describe the Chaum's Blind Signature Scheme. 5

SECTION-B (30 Marks)

5. a) Describe the PGP packets 3
b) Explain briefly the five headers of S/MIME that are added to the e-mail header. 3
c) What is the number of padding bits in case of SHA-512 if the length of the original message is 2590 bits? 2
d) Describe the Handshaking protocol in SSL. 2
6. a) In RSA, given $e = 13$ and $n = 133$, encrypt the message "CSE" using 00 to 25 for letters A to Z. 5
b) Discuss the differences between SSL and TLS. 3
c) In PGP, explain how Bob and Alice exchange the secret key for encrypting messages 2
7. a) Explain the security of secret voting scheme. 4
b) Explain a schematic diagram of a voting scheme. 6
8. a) What will be the transmitted frame if the given frame is 100110101 using the generator: $G(x) = x^3 + x^2 + 1$ when CRC code is used? 5
b) Find out the hamming code for "CSE". 5

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Course No.: CSE453

Full Marks: 60

N.B.:

Course Title: Digital Image Processing

Time: 3 (Three) Hours

- 1) Answer **SIX** questions taking any **THREE** from each section.
- 2) The figures in the right margin indicate full marks.
- 3) Sequence must be maintained for each question.

SECTION-A (30 Marks)

1. (a) Define Image. Suppose 'P' is a point at (100,100) position. Now find the coordinates of $N_4(P)$, $ND(P)$. How many bits required to store $M \times N$ size image with 'K' intensity level 4
- (b) What are the necessary conditions for a distance function? Write a short note on bilinear and bicubic interpolation. 4
- (c) What are the differences between Rods and Cones? 2
2. (a) Explain the digital image representation and their applications. 4
- (b) Consider the image segment shown below: 4

4	2	3	2(q)
3	3	1	3
2	3	2	2
(p) 2	1	2	3

Complete the length of shortest-4, shortest-8 and shortest-m paths between pixels p & q where, $v = \{1, 2\}$. If a particular path does not exist between these two points, explain why?

- (c) What do you mean by contrast stretching? 2

3. (a) Define histogram equalization. Consider the 5×5 , 3 bits/pixel image shown below: 4

0	5	7	7	5
7	2	6	2	6
6	1	4	7	0
6	3	1	7	6
1	3	2	4	4

- i) Perform histogram equalization to the image.
- ii) What will happen if we apply the same process on the resultant image of step (i).

- (b) Show the various techniques in frequency domain to enhance a image with necessary examples. 4
- (c) How is image processing related to computer vision? 2

4. (a) Let a 4-bit, 4×4 image passes through an intensity transformation function given by $s = T(r) = \alpha \log_2(1 + r) + \beta$, where α, β are constants. Few pixels are available in the input and output images, as shown below. 4

	3		3
9		15	B
	A		
1			3

$\rightarrow T(r) \rightarrow$

	5		5
C		11	9
3	8		
D	9		5

Input Image

Output Image

What are the values of the pixels A, B and C, D in the input and output images respectively?

- (b) State the steps of frequency domain filtering. Why do we need the centering of Fourier transform of an image? 4
- (c) What is mean filtering? 2

SECTION-B (30 Marks)

$D = 5$

5. (a) Consider the following 5x5 8-bit image with grayscale range of [0, 9].

0	5	7	7	5
7	2	6	2	6
6	9	7	7	0
6	6	1	7	6
9	3	2	4	8

Compress the image using Shannon-Fano encoding techniques and also discuss the efficiency of this compression.

- (b) What is histogram? Explain the histogram processing with example.

6. (a) What is image degradation and restoration? Explain them with example.

- (b) Explain inverse filtering process to restore a noisy image. What is the shortcoming of inverse filter? How does Wiener filter overcome such shortcoming?

- (c) What is the difference between Shannon-fano and Huffman coding?

7. (a) Define Pseudo color image processing. Discuss about intensity slicing with appropriate diagram.

- (b) Assuming continuous intensity values, derive an intensity transformation function that can make the intensity histogram uniform. Prove mathematically that the intensity histogram of the transformed image is uniform and independent of the intensity histogram of the original image.

- (c) For discrete intensity values, propose a transformation function for histogram equalization.

8. (a) Define image representation and description. Measure the entropy and uniformity of the following 3-bit image:

1	2	0	7	5
6	2	3	4	3
4	0	2	1	5
4	7	6	2	4
5	6	4	3	3
5	4	1	2	4

- (b) Define Hue and Saturation. Calculate the HSI value of the following RGB point: (155, 210, 75)

- (c) How can you apply a spatial filter on a full color image?

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Course No.: CSE465

Full Marks: 60

Course Title: Robotics

Time: 3 (Three) Hours

N.B.:

- 1) Answer **SIX** questions taking any **THREE** from each section.
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- 3) Sequence must be maintained for each question.

SECTION-A (30 Marks)

1. a) What is a Robot and Robotics? 2
b) Draw a simple robotic system using the basic components. 4
c) What do you mean by DOF? Write the names of the lower pair connectors of 1, 2 and 3 DOF. 4
2. a) Why is the coordinate system used in a robotic system? Explain the cylindrical, spherical and revolute coordinate system with respect to joint pairs. 6
b) What do you mean by the work envelope of a robotic system? 2
c) What do you mean by forward and backward robot kinematics? 2
3. a) Program DEMO. A 3

```
1. APPRO PART, 50
2. MOVES PART
3. CLOSEI
4. DEPARTS 150
5. APPROX BOX, 200
6. MOVE BOX
7. OPENI
8. DEPART 75
END
```
- b) Explain the program
- c) A robot that separates different types of garbage from mixed garbage. Write your design considerations. 4
d) What are the challenges of designing modern robots? 3
4. a) Define actuators? Explain the working principles of different kinds of actuators. 5
b) Draw a simple positional servo system. 5

SECTION-B (30 Marks)

5. a) Why is a stepper motor used in a robotic system? Describe the operating principle of a stepper motor. 6
b) In a stepper motor what factors are responsible for measuring each step size? 2
c) Draw a cross section of a stepper motor whose step size is 18 degrees. 2
6. a) What is a sensor? Write five key characteristics of a sensor. 3
b) How does artificial skin work? 3
c) Describe the working mechanism of a proximity sensor. 4
7. a) State the three laws that a robot must obey. 3
b) Explain the low- and high-level vision of a robot. 4
c) How can you program for a modern robot? 3
8. a) Write the role of a microcontroller to operate a robot. 3
b) Could a robot be lefty or righty?
c) Differentiate between contact and non-contact sensor.
d) How does a tactile array sensor work?

Bangabandhu Sheikh Mujibur Rahman Science and Technology University
Department of Computer Science and Engineering (ICT)
4th Year 2nd Semester B.Sc. Engg. Examination-2022 Session: 2018-19

Course No.: CSE 455

Course Title: Computer Peripherals and Interfacing

Full Marks: 60

Time: 3 (Three) Hours

N.B.:

- 1) Answer **SIX** questions taking any **THREE** from each section.
- 2) The figures in the right margin indicate full marks.
- 3) Sequence must be maintained for each question.

SECTION-A (30 Marks)

1. a) If peripherals could operate at similar speeds to CPU, peripheral adapter would no longer be needed for the operation. True or False? Justify your answer. 5
b) What is an interrupt service routine? Explain the working principle of interrupt controller. 5
2. a) Define key ghosting and key rollover. What are the differences between mechanical and membrane keyboards in terms of structure and performance? Briefly explain. 6
b) How do specialized keyboards improve user experience? Also, Discuss the significance of DPI (dots per inch) in the performance of a mouse. 4
3. a) Define aspect ratio, viewable area, and resolution of computer display monitor. Explain how the resolution of a monitor affect output quality. 4
b) Differentiate between LCD and LED monitors in terms of technology and picture quality. 3
c) Define OMR, OCR, ICR, MICR. Also explain the purpose and functionality of OMR technology. 3
4. a) What is barcode? Discuss the role of barcode scanners in inventory management and retail operations. Provide examples of its advantages over manual data entry. 4
b) With the rise of digital platforms and e-books, some argue that ISBNs are becoming less relevant in modern publishing. Discuss the relevance of ISBNs in the digital age. 3
c) The first nine digits of the ISBN-10 of the sixth edition of a book are 007288008. What is the check digit? Is 084930149X a valid ISBN-10? 3

SECTION-B (30 Marks)

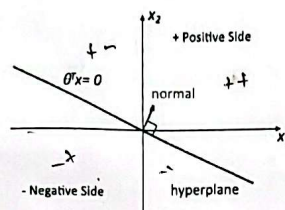
5. a) Define interfacing in the context of computer systems and explain its importance in hardware-software interaction. 4
b) What is the difference between memory-mapped I/O and port-mapped I/O? Provide examples of when each is used. 3
c) What is chipset? What is the difference between the Northbridge and Southbridge components of a traditional chipset architecture? 3
6. a) What is interrupt? Explain the purpose of interrupt in computer organization. Also, discuss the concept of interrupt priority and how it is managed in a multi-interrupt system. 4
b) Interrupts are often considered better than polling for handling input/output tasks in computers. Do you agree or disagree with this statement? Also, provide your opinion on which method is more suitable for general use, with reasons. 4
c) Write down the difference between physical port and virtual ports. Why are they important for device communication? 2
7. a) Briefly define the components of the Peripheral Adapter with a diagram. 5
b) Describe DMA works with diagrams. 5
8. a) What is system bus? Draw the block diagram of system bus and explain each component in detail. 4
b) Explain how the speed of a system bus affects overall system performance, focusing on the impact of data transfer rates. Also write down the advantages of AGP over PCI bus. 3
c) SATA and SCSI are two different interfaces used for connecting storage devices to a computer. Which interface do you think is better for modern storage needs? Justify your answer. 3

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SECTION-A (30 Marks)

- What is machine learning? Why is Machine Learning getting so much attention recently? 1+2
 - Distinguish between traditional/rule-based programming and machine learning with their workflow diagram. 4
 - Google Photos often automatically creates albums based on common themes or events in your photos, such as vacations or birthdays. What type of machine learning technique is used for this feature and why? 3



- In two dimensions, $\theta = [\theta_1, \theta_2]$ can define a hyperplane. Let $\theta = [1, 2]$. Give a vector that lies on the hyperplane given by the set of all $x \in \mathbb{R}^2$ such that $\theta^T x = 0$
 - Using the same hyperplane, determine a vector that is normal to the hyperplane.
- b) Consider a linear classifier through the origin in 4 dimensions, specified by $\theta = (1, -1, 2, -3)$ 6
Which of the following points x are classified as positive, i.e. $h(x; \theta) = +1$? Give explanation.
- $(1, -1, 2, -3)$
 - $(1, 2, 3, 4)$
 - $(-1, -1, -1, -1)$
 - $(1, 1, 1, 1)$

$$x_1 + x_2 + 2x_3 + 3x_4 = 0$$

$$1 + 1 + 4 + 9 = 15 \neq 0$$

Consider another parameter vector $\theta' = (-1, 1, -2, 3)$ Does θ' represent the same hyperplane as θ does? Does θ' represent the same classifier as θ does? Give explanation

- Which of the following pictures illustrates the squared loss metric? Assume that the dark line is described by θ , θ_0 , the black dots are the (x, y) data, and the light lines indicate the errors. 3

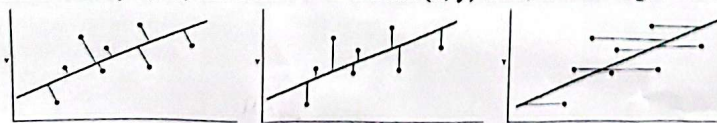


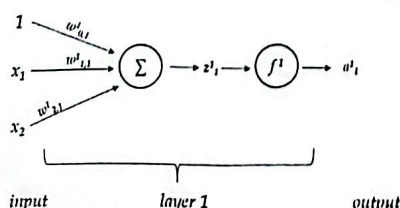
Fig. a

Fig. b

Fig. c

- Now, we will consider the classification of a different set of X and Y data, with a different single layer network having the following structure and activation function: 7

$$z = w_{1,1}^1 x_1 + w_{2,1}^1 x_2 + w_{0,1}^1$$



Assume the initial weights are $w_{0,1}^1 = 1$, $w_{1,1}^1 = 1$, $w_{2,1}^1 = 1$ the step size is 0.5.
The current training example is $x^{(i)} = [1, 2]^T$, $y^{(i)} = -1$.
What is the output value of $a_1^{(i)}$, given current input $x^{(i)}$ and the current weights?